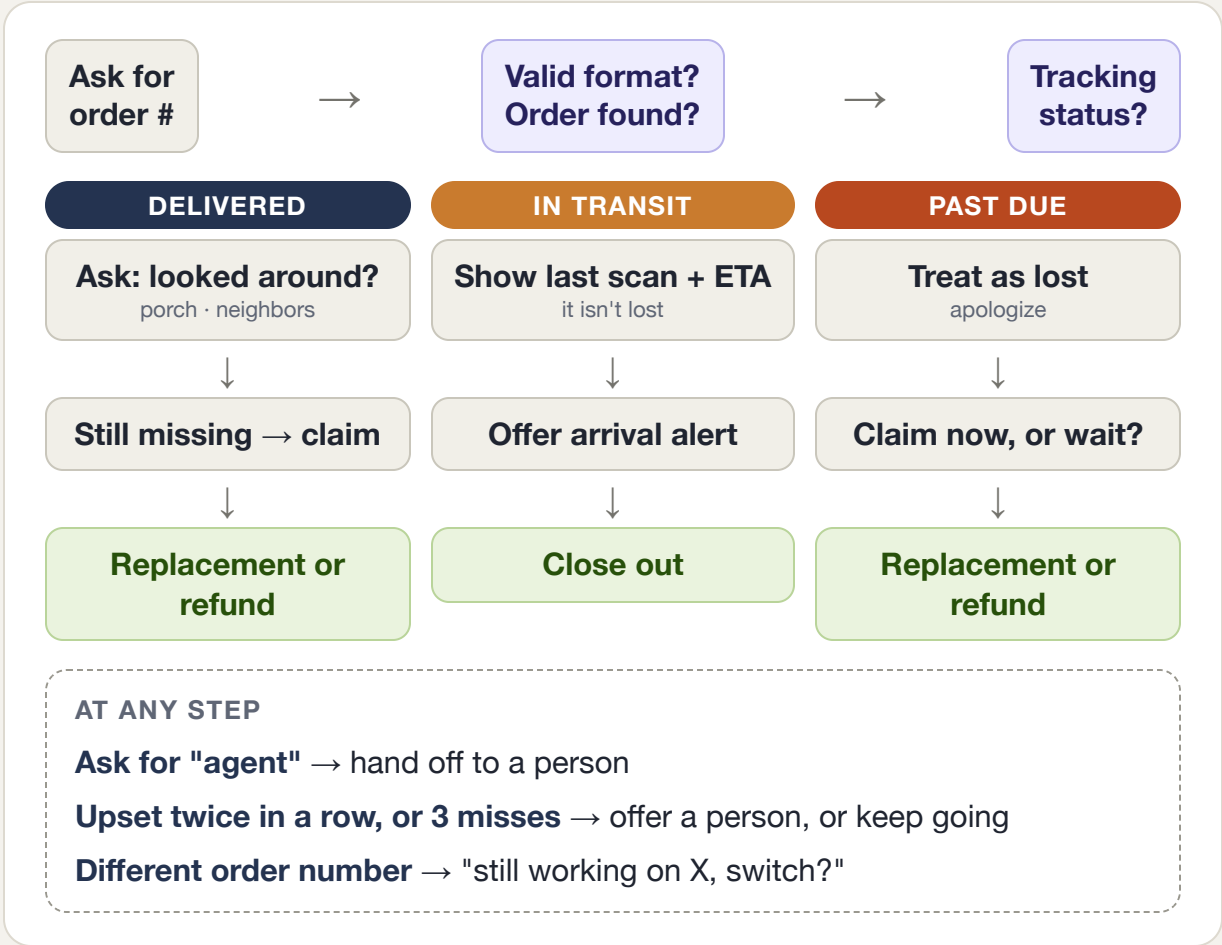


Work out what's really wrong first, then act

- 1 **Triage before anything else.** "Lost package" is usually one of three situations, and only one is actually lost: *delivered but missing*, *still in transit*, or *past due with no movement*. Each gets a different path.
- 2 **A designed flow with natural language on top.** Customers type however they talk. The steps are fixed, so every path is predictable and testable.
- 3 **Errors handled by kind.** A bad format and an unknown order get different messages. Three unmatched messages, or two upset ones, offer a person. A new order number mid-chat asks before switching.
- 4 **Never leave the customer hanging.** Every reply ends by asking what they want and naming the options.



Gemini decides and words it. The app holds the facts.

Browser · HTML / CSS / vanilla JS chat UI
Sends only the customer's text and a session id

↓ text only

Node + Express server · holds conversation state and the API key
Guardrails, rate limits, CORS, fallbacks

↓ each turn

engine.js · the flow
9 states, 18 moves, the facts, the draft wording

Gemini (function calling)
① pick one legal move ② reword the draft

- **Only legal moves are offered** to Gemini each step, and it must call exactly one. Illegal picks are rejected.
- **The reword is checked.** If it changes a number, date or order id, or drops the follow-up question, the plain draft is sent.
- **Graceful fallbacks.** Gemini down → keyword rules. Server down → the same engine runs in the browser.
- **98 automated tests** cover every order, error path, edge case and guardrail. Deployed free on Render.

this is so annoying, it's been weeks
→ **unclear** · gemini · upset

I completely understand your frustration, and I'm really sorry you've had to wait this long. Would you like me to file the claim now, keep watching it for a few more days, or get you an agent?

SERIOUSLY?? this is ridiculous
→ **unclear** · gemini · upset

I completely get why you're upset, and I'm really sorry about this mess. Would you like me to file the claim right away, keep watching it for a few more days, or connect you with an agent?

I hear you, and I am so sorry for how frustrating this whole situation is. Would you like me to connect you with a person, or would you prefer to keep going here with me?

Yes, get me an agent No, let's keep going

Type a message... Send

Agent mode: each customer message shows the move chosen, who chose it, and an upset flag. Two upset in a row offers a person.

Making an AI natural without making it unsafe

Letting an LLM talk without letting it promise things

Split the jobs. The app owns state and facts; Gemini only picks a legal move and rewords a draft. Rewrites that change a fact are thrown away, and a manipulated model can only pick the wrong move from a short list.

Sounding human, but never leaving the customer stuck

Tuned by testing real chats. Early replies ended on statements, and frustration alone handed off to a person. Now every reply asks what to do next, and a person is offered only when asked, after 3 misses, or after 2 upset messages.

Messy real-world inputs

20 mock orders covering 11 statuses exposed gaps: a customer pasting a new order number mid-chat was "unclear", and a wrong-address delivery was told to check the door. Now the bot confirms before switching, and misdelivered orders go straight to a claim.

Hosting and connecting a free server

CORS, cold starts and file:// pages. A page opened from disk has a null origin, and the free Render tier sleeps for 30 to 60 seconds. The page falls back to keyword mode so the demo never breaks, and the debug panel explains why it fell back.

From prototype to something customers can rely on

Make it real

- **Real carrier lookups** (USPS, UPS, FedEx) in place of the mock orders.
- **A true handoff.** Open a ticket or live chat and pass the full transcript so customers never repeat themselves.
- **Claims and alerts that actually happen:** file the claim, send the email or SMS.
- **Persistent sessions** so a restart or a sleeping server doesn't lose the chat.

Make it safer and smarter

- **Check meaning, not just numbers,** when validating Gemini's rewording.
- **An evaluation set** of messy real phrasings to score moves before each change.
- **Production hardening:** tighter CORS, authentication, and per-user rate limits.
- **More scenarios and languages** on the same flow engine.

WHAT I'D MEASURE

Where customers drop out of the flow, how often the bot resolves without a person, how often it falls back to keyword rules, and how often Gemini's rewording is rejected.